

Dual-Manifold Alignment Contrastive Learning for Knee Osteoarthritis Prediction

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Abstract—Research into knee osteoarthritis (OA) progression has gained significant interest as medical trends shift from disease treatment to prevention. Recent analysis models employ generative techniques or Explainable Artificial Intelligence (XAI) to quantify progression risk from radiographs. However, such methods typically assume that all OA progression is homogeneous, treating deterioration as a deterministic route from one state to another. This assumption fails to hold clinically, as progression pathways are often unique to each patient due to the disease’s complexity. To address this, we propose a novel framework, Dual-Manifold Alignment Contrastive Learning (DuMA-CLR). Our architecture utilizes a dual-head projector to map an input X-ray into distinct ‘Present’ and ‘Future’ latent manifolds. The DuMA-CLR objective then enforces geometric alignment between these time-points for stable patients, while strictly imposing trajectory divergence for progressing patients. This approach effectively estimates progression risk by preserving the heterogeneous structural changes unique to each subject, rather than collapsing them into a uniform class. Experimental results demonstrate that our method achieves higher Area Under Curve (AUC) scores while exhibiting greater robustness to multi-center data variations compared to state-of-the-art (SOTA) methods.

Index Terms—Medical AI, Osteoarthritis, Contrastive Learning

I. INTRODUCTION

Knee Osteoarthritis (OA) is a leading cause of disability worldwide, affecting nearly 600 million individuals [1], [2], causing burden on healthcare facilities and on the patients’ quality of life. As medical paradigms shift from late-stage treatment to early prevention, research for the ability to assess risk of knee OA progression grows increasingly prevalent. Furthermore, accurate risk estimation is critical not only for early interventions but also for clinical trial enrichment, where participants are screened for pharmaceutical trials to ensure that enrolled subjects are likely to demonstrate measurable progression within the study window [3], [4]. However, despite the availability of large-scale longitudinal datasets such as the Osteoarthritis Initiative (OAI) [5], predicting the trajectory of OA remains a formidable challenge due to the complex, multifactorial nature of the disease.

Recent advancements in deep learning address this challenge using diverse methodologies. Standard Convolutional Neural Networks (CNNs) have been widely developed to

classify future Kellgren-Lawrence (KL) grades [6] directly, while more recent approaches have utilized Denoising Diffusion Probabilistic Models (DDPM) [7] to visualize future disease states by generating predictive radiographs [8]. Additionally, research in Explainable Artificial Intelligence (XAI) has focused on highlighting saliency maps to interpret the morphological features associated with severity. While these methods have improved predictive performance, they typically rely on a fundamental, yet often overlooked, simplification: the homogeneity assumption. Most existing models treat disease progression as a uniform class, assuming that all patients who deteriorate follow a similar visual or latent trajectory from one state to another.

However, this assumption does not hold true clinically. OA progression is inherently heterogeneous [3]. In contrast to the non-progressiveness of knees with low risk estimation, progressing knees deteriorate through highly variable pathways—ranging from medial joint space narrowing and osteophyte formation to subchondral sclerosis. Forcing these diverse progression patterns into a learnable flow or direction only generalizes the disease progression as a deterministic vector from one state to another. Furthermore, standard contrastive or classification losses struggle to capture the granular, instance-specific features necessary for accurate forecasting. This phenomenon mirrors the Anna Karenina principle: stable patients are all alike, but every progressing patient progresses in their own way [9].

To address this limitation, we propose a novel framework, Dual-Manifold Alignment Contrastive Learning (DuMA-CLR). Unlike traditional methods that project features into a single static space, our architecture processes a single baseline radiograph and utilizes a dual-head projector to map the visual features into distinct ‘Present’ and ‘Future’ latent manifolds, supervised by their respective time-point KL grades. We introduce an asymmetric contrastive objective between the dual-head projections that governs the relationship between these spaces: it enforces strict geometric alignment between the predicted current and future states for stable patients (ensuring temporal consistency), while explicitly imposing trajectory divergence for progressing patients. By pushing progressing instances away from their baseline state and from each other

in the latent space, DuMA-CLR preserves the unique structural deviations of each subject rather than collapsing them into a uniform class.

Our main contributions are summarized as follows:

- 1) We propose DuMA-CLR, a novel contrastive learning framework that processes single baseline radiographs to explicitly model the heterogeneity of OA progression, avoiding the limitations of symmetric clustering assumptions.
- 2) We introduce a Dual-Manifold strategy where a dual-head projector maps a single visual input into distinct 'Present' and 'Future' latent spaces. These spaces are aligned via an asymmetric objective that enforces temporal consistency for stable patients and trajectory divergence for progressors.
- 3) We demonstrate through extensive experiments on the OAI dataset that our method achieves state-of-the-art (SOTA) AUC scores and exhibits superior robustness to domain shifts across different data acquisition sites compared to existing baseline methods.

II. METHODOLOGY

A. Problem Formulation

To begin, we define the task of forecasting knee OA progression. Let $\mathcal{D} = \{(x_i, y_i^t, y_i^{t+\Delta})\}_{i=1}^N$ represent the dataset, where x_i denotes the input radiograph at baseline time t . The labels y_i^t and $y_i^{t+\Delta}$ represent the KL-Grades at baseline and at the 48-month future time point, respectively.

The final output, progression, is defined as a binary outcome p_i , where $p_i = 1$ if $y_i^{t+\Delta} > y_i^t$ and $p_i = 0$ otherwise. While standard approaches typically utilize this label to supervise a binary classification objective by forcing all samples with $p_i = 1$ into a single cluster, such a formulation simplifies the heterogeneous nature of the disease. Furthermore, such simplification does not take into account the implicit relationship between p_i and the labels y_i^t and $y_i^{t+\Delta}$. In our framework, we utilize p_i not as a target for unification, but as a signal to enforce geometric properties: specifically, $p_i = 0$ triggers manifold alignment, while $p_i = 1$ triggers trajectory divergence to capture the diverse magnitude of structural deterioration.

B. Preliminaries: Probability Risk Estimation

Heterogeneity for the prediction of progression can be improved by re-framing the solution as a secondary output to primary results rather than being a primary output itself. While recent methods employed progression transitions as a deterministic primary output from one grade to another [8], it outputs the final prediction score as a secondary result to the main classification problem. A diagram of this secondary result approach is as shown in Fig. 1.

The probabilistic risk estimation architecture involves classifying the current KL-Grade, $\hat{y}^t$, and the expected future KL-Grade, $\hat{y}^{t+\Delta}$ first. Then, following the principle that KL-grades can only worsen overtime, all softmax values of $\hat{y}^{t+\Delta}$, which labels are higher than $\hat{y}^t$, are treated as progression

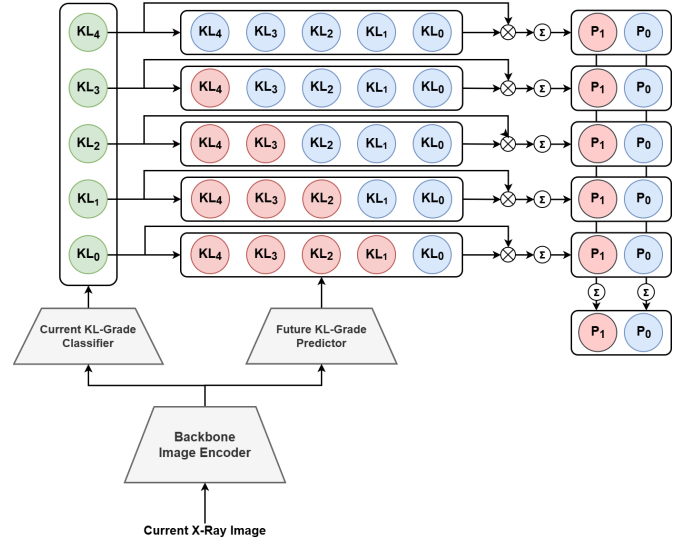


Fig. 1. Illustration of probabilistic risk estimation, where risk is framed as a secondary output to the primary classification results. Rather than treating progression as a deterministic primary output from one grade to another, the final risk prediction score is derived secondarily through the probabilistic summation of the main classification outputs.

probabilities. The progression probabilities are multiplied with the softmax value of $\hat{y}^{t+\Delta}$, which are all then added together to get a final risk prediction score in the form of a probability.

In this work, we retain this secondary probabilistic structure for the final risk calculation. Nevertheless, the baseline remains dependent on mapping deterministic status transitions as its primary task. DuMA-CLR aims to adjust this primary stage by enforcing geometric divergence, ensuring that the prediction of change emerges as a resulting effect of structural separation rather than a rigid classification output.

C. DuMA-CLR Architecture

In this section, we introduce the Dual-Manifold Alignment Contrastive Learning (DuMA-CLR) framework. As shown in Fig. 2, our model consists of three main components: a shared backbone encoder, a dual-head projection module for both supervised contrastive learning and classification, and the final DuMA alignment objective.

1) *Shared Backbone Encoder*: We utilize a ConvNeXt-Base architecture [10] as the feature extractor due to its ability to capture both local texture details and global shape semantics. Given an input image x_i , the encoder $f(\cdot)$ outputs a high-dimensional feature vector $h_i = f(x_i) \in \mathbb{R}^d$. Consistent with standard contrastive learning protocols [11], we utilize this vector as the input for our projection modules, decoupling the learned representation from the specific downstream tasks.

2) *Dual-Head Projector*: To learn both current and future representations effectively, we employ two distinct Multi-Layer Perceptron (MLP) heads responsible for mapping h_i into independent spaces:

$$\begin{aligned} z_{curr} &= p_{curr}(h_i) \\ z_{fut} &= p_{fut}(h_i) \end{aligned} \quad (1)$$

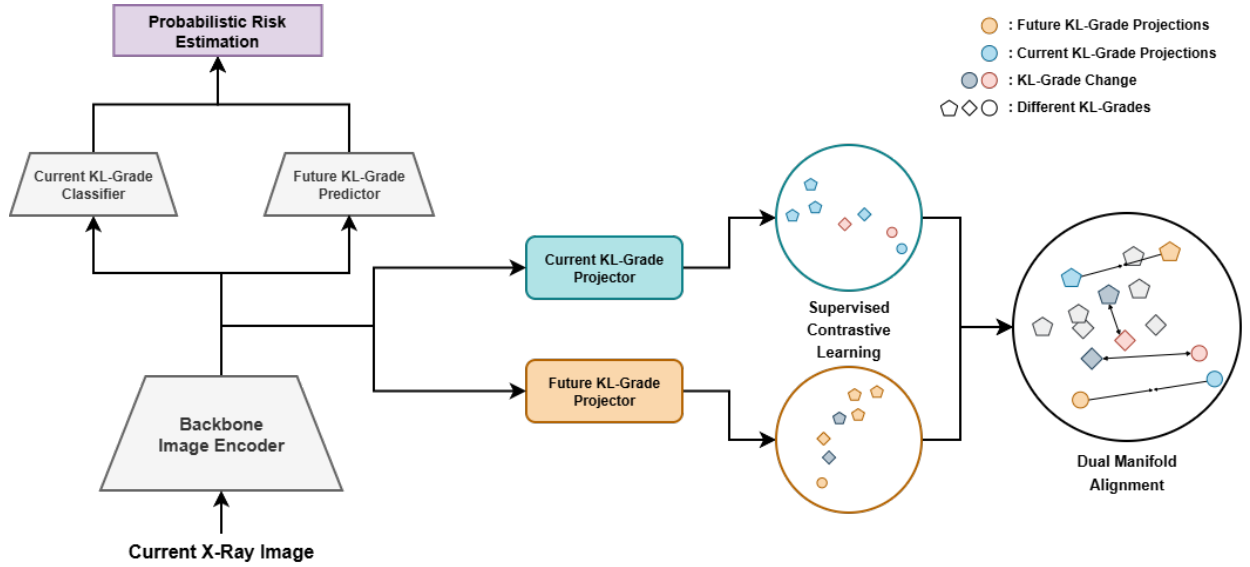


Fig. 2. Overview of the proposed DuMA-CLR framework. A shared backbone encoder extracts features from the input X-ray. These features are input into two parallel branches: one for classifying the current and future KL-grades, and another for projecting the features into distinct 'Current' and 'Future' latent manifolds via supervised contrastive learning. The final Dual-Manifold Alignment step enforces geometric consistency for stable patients, while maximizing divergence for progressing patients, capturing the heterogeneity of disease progression.

Where p_{curr} and p_{fut} each refer to the projector heads for current KL-Grades and future KL-Grades, respectively. The outputs z_{curr} and z_{fut} are used to conduct supervised contrastive learning, where data with the same KL-Grades are pulled towards each other, while data with different grades are pushed against each other in the latent space. p_{curr} is supervised on the labels y_i^t and p_{fut} is supervised on the labels $y_i^{t+\Delta}$, ensuring that the two different projection spaces will be independent from each other whilst learning a latent space tailored for its temporal position.

D. DuMA-CLR Objective

The learning objective of our framework is twofold: first, to establish discriminative latent structures within each independent time-point, and second, to enforce geometric constraints between these time-points.

As described in the previous section, the intra-manifold structure is learned via Supervised Contrastive Learning losses ($\mathcal{L}_{sup}$) applied independently to z_{curr} and z_{fut} . To govern the inter-manifold relationship, we introduce the DuMA objectives. We define the pairwise cosine similarity between the current and future representations as:

$$\text{sim}(z_{curr}, z_{fut}) = \frac{z_{curr}^\top z_{fut}}{\|z_{curr}\| \|z_{fut}\|} \quad (2)$$

1) *Manifold Alignment for Stable Patients:* For patients who do not experience progression ($y^t = y^{t+\Delta}$), the anatomical state encoded in the future manifold should be identical to that of the current manifold. Therefore, we enforce alignment between the two embeddings by maximizing the similarity between them. The alignment loss $\mathcal{L}_{align}$ is defined as:

$$\mathcal{L}_{align} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \left(1 - \text{sim}(z_{curr}^{(i)}, z_{fut}^{(i)}) \right) \quad (3)$$

where $\mathcal{S}$ denotes the set of stable patients. This term pulls the future embedding towards the current embedding, enforcing representational consistency between the two latent spaces.

2) *Trajectory Divergence for Progressing Patients:* For patients who experience progression ($y^{t+\Delta} > y^t$), the future state must be distinct from the current state. Furthermore, consistent with the definition that progression pathways are unique, we enforce divergence between the time points. The divergence loss $\mathcal{L}_{div}$ is defined as the direct minimization of the pairwise cosine similarity:

$$\mathcal{L}_{div} = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \text{sim}(z_{curr}^{(i)}, z_{fut}^{(i)}) \quad (4)$$

where $\mathcal{P}$ denotes the set of progressing patients. By minimizing this term, the model is penalized for alignment between the current and future embeddings, effectively pushing the future representation away from the baseline state to model the drift of the disease.

3) *Total Objective:* The final objective function combines the intra-manifold supervised contrastive losses with the inter-manifold geometric alignment terms:

$$\mathcal{L}_{total} = (\mathcal{L}_{sup}^{curr} + \mathcal{L}_{sup}^{fut}) + \lambda_{align} \mathcal{L}_{align} + \lambda_{div} \mathcal{L}_{div} \quad (5)$$

where $\mathcal{L}_{sup}^{curr}$ and $\mathcal{L}_{sup}^{fut}$ represent the supervised contrastive losses for the current and future projection heads, respectively.

In essence, this formulation creates a competitive equilibrium between the objectives. The Supervised Contrastive Loss ($\mathcal{L}_{sup}$) encourages the model to learn discriminative features independent to each temporal latent space, whereas the DuMA-CLR objectives ($\mathcal{L}_{align}, \mathcal{L}_{div}$) enforce geometric dependencies between them. Crucially, the key distinction lies in the treatment of progressing patients: while stable

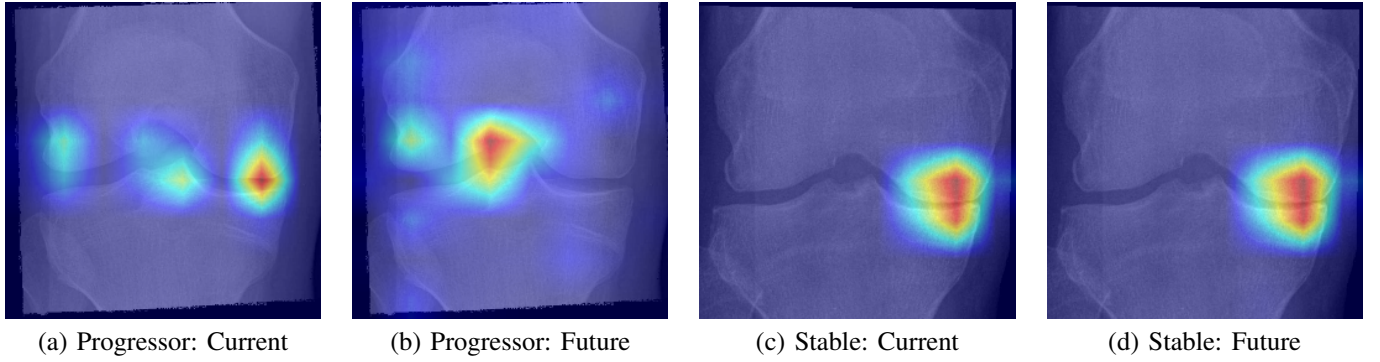


Fig. 3. Grad-CAM visualization of the attention maps generated by the dual projection heads of DuMA-CLR. (a) and (b) display the attention maps from the current and future KL-Grade classification heads for a progressing patient. (c) and (d) display the attention maps from the current and future KL-Grade classification heads for a stable patient.

representations are harmonized across time, progressing representations are geometrically forced to break away from their baseline identity. Consequently, the final learned manifold becomes semantically meaningful for both the baseline and 48-month states, effectively characterizing progression not as a simple label switch, but as a measurable divergence from the stable norm.

III. EXPERIMENTS

A. Dataset and Implementation Details

1) *Dataset*: For this paper, we focused on the OAI dataset [5], a large-scale longitudinal study of knee OA. To rigorously evaluate the prognostic capabilities of our model, we utilized baseline radiographs (Month 00) as the sole visual input. Supervision was provided by extracting the ground truth KL-Grades for both the baseline image and the corresponding patient’s follow-up status at the 48-month (4-year) time point. The dataset was filtered to exclude patients with missing follow-up data or total knee replacements prior to the endpoint. This resulted in a total of 4,796 image pairs. The images were cropped to the knee joint region and resized to 512×512 pixels. Horizontal flipping was applied to X-Ray images of the left knee to maintain positional semantic consistency.

To ensure the model’s robustness to different clinical environments, the dataset was split based on the data acquisition site rather than a random shuffle. Specifically, data acquired from Site D was reserved for the validation and test sets, which comprised approximately 25% of the total dataset. This resulted in an approximate train-validation-test split ratio of 6:1:1, allowing for the evaluation of the model’s robustness on unseen institutional data distributions.

2) *Training Setup*: All models were trained using the AdamW optimizer [12] on 4 NVIDIA Tesla V100 GPUs. We used a batch size of 64 and a learning rate of 1×10^{-4} with a cosine annealing scheduler. The hyperparameters λ_{align} and λ_{div} were empirically set to 0.5 and 1.0, respectively.

B. Quantitative Results

To evaluate the effectiveness of DuMA-CLR, we compared it against several state-of-the-art baselines on the OAI dataset.

The first chosen baseline, Multi-task Classification Model [13], represents the state-of-the-art performance for MLP only approaches, whilst the second baseline, Risk Estimation [8], represents the state-of-the-art performance for prediction through generative means. We utilized Area Under the Curve (AUC) as our primary metric for risk estimation performance.

As summarized in Table I, standard classification approaches such as the Multi-task Classification Model achieved an AUC of 0.6870 [13]. The recent Risk Estimation method, which serves as a strong state-of-the-art baseline, achieved a competitive AUC of 0.7289.

Our proposed DuMA-CLR framework outperformed both baselines, achieving a state-of-the-art AUC of 0.7638. This represents a substantial +4.79% improvement over the strongest baseline. This performance gain is statistically significant and indicates that explicitly modeling the divergence of progressing trajectories allows the network to identify high-risk patients that standard symmetric models miss.

TABLE I
COMPARATIVE ANALYSIS OF RISK ESTIMATION PERFORMANCE (AUC)

Models	AUC $\uparrow$
Multi-task Classification Model	0.6870
Risk Estimation (SOTA)	0.7289
DuMA-CLR (Proposed)	0.7638

C. Qualitative Analysis

To interpret the internal mechanics of DuMA-CLR, we visualized the attention maps of the dual-head projector using Gradient-weighted Class Activation Mapping (Grad-CAM) [14]. As hypothesized, the geometric constraints of our loss function are directly reflected in the model’s focus regions (Fig. 3).

For stable patients, the attention maps between the two heads are visually consistent. This confirms that the Manifold Alignment term successfully enforces temporal consistency, teaching the model that the current and future states of a stable knee are represented equivalently.

In contrast, a significant discrepancy can be observed for patients with progression. While one head focuses on the

general joint structure, the corresponding future/divergent head shifts its attention to specific pathological indicators such as the medial joint boundary. This visual difference demonstrates the effectiveness of the Trajectory Divergence term. The model actively learned that the future state of a progressor is distinct from its current state, forcing the latent representations to diverge.

D. Ablation Study

To better understand the impact of the proposed architectural modifications, we conducted an ablation study (Table II).

TABLE II
ABLATION STUDY OF DuMA-CLR

Configuration	AUC	Drop
DuMA-CLR (Full Model)	0.7638	-
w/o Future Head Projector	0.6983	-8.58%
w/o Divergence Loss ($\mathcal{L}_{div}$)	0.6936	-9.19%

First, we evaluated the importance of the Future Head Projector by removing the branch responsible for predicting the future latent state. This resulted in a performance drop of 8.58% (AUC 0.6983), confirming that explicitly modeling a distinct latent space with temporal displacements, rather than relying on only the current state latent space, is essential for accurate forecasting.

Additionally, removing the Divergence Loss ($\mathcal{L}_{div}$) resulted in the most significant performance degradation, with a drop of 9.19% (AUC 0.6936). Without this asymmetric constraint, the model reverts to standard symmetric contrastive learning, which forces all progression patterns into a single cluster. The sharp decline in AUC provides strong empirical evidence that the symmetric assumption is detrimental to performance, and that enforcing geometric divergence is the primary driver of DuMA-CLR’s success.

IV. CONCLUSION

In this paper, we proposed DuMA-CLR, a novel framework that addresses the homogeneity assumption prevalent in existing OA risk estimation models. By leveraging a dual-manifold projection strategy and an asymmetric contrastive objective, our method successfully disentangles stable and progressing trajectories. Our experimental results on the OAI dataset confirm the validity of this approach, achieving a state-of-the-art AUC of 0.7638 and outperforming existing baselines by over 4%.

However, while our results represent a significant improvement, the current AUC of roughly 76% indicates that the model has not yet reached the clinical ideal of 80% required for robust deployment. We hypothesize that this limitation stems from the unconstrained nature of our divergence loss: while DuMA-CLR successfully learns that progressing representations must lie outside the stable norm, it does not explicitly enforce a specific directional pathway for this divergence. Consequently, the model captures the magnitude of deviation

but may struggle to characterize the distinct morphological semantics of that change.

Future work will address this by integrating a generative guidance module into the dual-manifold framework. Specifically, we aim to utilize latent diffusion models conditioned on the estimated risk value to guide the divergent embeddings. Rather than forcing a deterministic linear path, this generative component will define a non-deterministic cone of progression, providing the necessary structural guidance to refine the trajectory while preserving the heterogeneous nature of the disease.

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